VBF-T1R1NOFORCE-BOM-01 - Bill of Materials (g/cell and kg/kWh)

BOM

Component	Mass (g/cell)	kg/kWh	Source note
Positive active material (NMC811)	16.168	0.7898	96 wt% of positive layer 16.842 g (literature default split; parameter set lacks additive/binder keys)
Positive conductive additive (carbon)	0.337	0.0165	2 wt% literature default
Positive binder (PVDF)	0.337	0.0165	2 wt% literature default
Negative active material (graphite)	11.64	0.5686	96 wt% of negative layer 12.125 g (literature default split)
Negative conductive additive (carbon)	0.243	0.0119	2 wt% literature default
Negative binder	0.243	0.0119	2 wt% literature default
Separator (PP 12 um, porosity 0.47)	0.259	0.0127	layer_kg_m2 0.00252492 x 0.1027 m2 (calc-energy r6_c7_energy.json)
Electrolyte (EC/EMC + LiPF6)	6.743	0.3294	pore volume 5.6193e-6 m3 x 1200 kg/m3 (literature density, fill factor 1.0; parameter set lacks electrolyte density)
Positive current collector (Al 16 um)	4.437	0.2167	layer_kg_m2 0.043200 x 0.1027 m2 (calc-energy)
Negative current collector (Cu 12 um)	11.042	0.5394	layer_kg_m2 0.107520 x 0.1027 m2 (calc-energy)
Enclosure / tabs	Not modeled		Not modeled
TOTAL (incl. electrolyte estimate)	51.449	2.5132	mechanical sum; contract mass (electrolyte excluded) = 44.705 g from r6_c7_energy.json:mass_kg
Cell energy (kWh)	0.02047186640547461		r6_c7_energy.json:energy_wh / 1000